REPO RADAR

Linux: from a Helsinki hobby to the world's kernel

The story in 6 facts

1. **25 August 1991**: Linus Torvalds, a 21-year-old computer science student at the University of Helsinki, posts on the `comp.os.minix` newsgroup: "I'm doing a (free) operating system (just a hobby, won't be big and professional like gnu) for 386(486) AT clones."
2. **Why he started**: students learned on MINIX (Andrew S. Tanenbaum, Vrije Universiteit Amsterdam, 1987). Its source code was available, but modification and redistribution were restricted until MINIX moved to a BSD licence in April 2000.
3. **17 September 1991**: version 0.01 is uploaded to the FUNET FTP server (`ftp.funet.fi`). The kernel is about 10,000 lines of code.
4. **The name**: Torvalds wanted "Freax". Ari Lemmke, who ran the server, named the project "Linux" on the server.
5. **2005 to 2017**: about 15,600 developers from more than 1,400 companies contributed to the kernel (Linux Foundation, 2017 report).
6. **November 2017**: all 500 systems on the TOP500 list of the fastest supercomputers ran Linux.

Where the code lives

- Official home: kernel.org
- Read-only GitHub mirror of Linus Torvalds' tree: github.com/torvalds/linux (pull requests are not accepted there; patches go through the mailing lists).

Look at it yourself (read-only, safe)

```
git clone --depth 1 https://github.com/torvalds/linux.git
cd linux
git log --oneline -5
ls
```

`--depth 1` downloads only the latest snapshot, not 30+ years of history.

3 things to explore

- `README` and `Documentation/process/submitting-patches.rst`: how a patch reaches the kernel.
- `MAINTAINERS`: who looks after each part of the code.
- `kernel/` and `init/main.c`: where the kernel starts.

Sources

- History of Linux: https://en.wikipedia.org/wiki/History_of_Linux
- MINIX: https://en.wikipedia.org/wiki/Minix
- Linux Foundation, 2017 Linux Kernel Report: https://www.linuxfoundation.org/blog/2017-linux-kernel-report-highlights-developers-roles-accelerating-pace-change/
- Linux Foundation, Linux runs all of the world's fastest supercomputers (2017): https://www.linuxfoundation.org/blog/blog/linux-runs-all-of-the-worlds-fastest-supercomputers

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